Graid Technology Inc.

The Best HPC Cluster Solution Drive by SupremeRaid with BeeGFS

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BETTER TOGETHER

Why BeeGFS + SupremeRAID™?

High Performance

SupremeStorre turnkey solution delivers extremely high performance with **80 GB/s** and lowest lantency.

Body Mirror by BeeGFS and Cross Node HA by SupremeRaid

Provides high availability, ensuring data access and business continuity even during partial system failures.

Boundless Scaling out

With BeeGFS advantages, scales out storage and performance by adding more nodes, allowing for linear increases in both capacity and throughput.

Adapt to Diverse Workloads

Whether facing high-density data processing tasks or random I/O operations, our system ensures efficient performance.

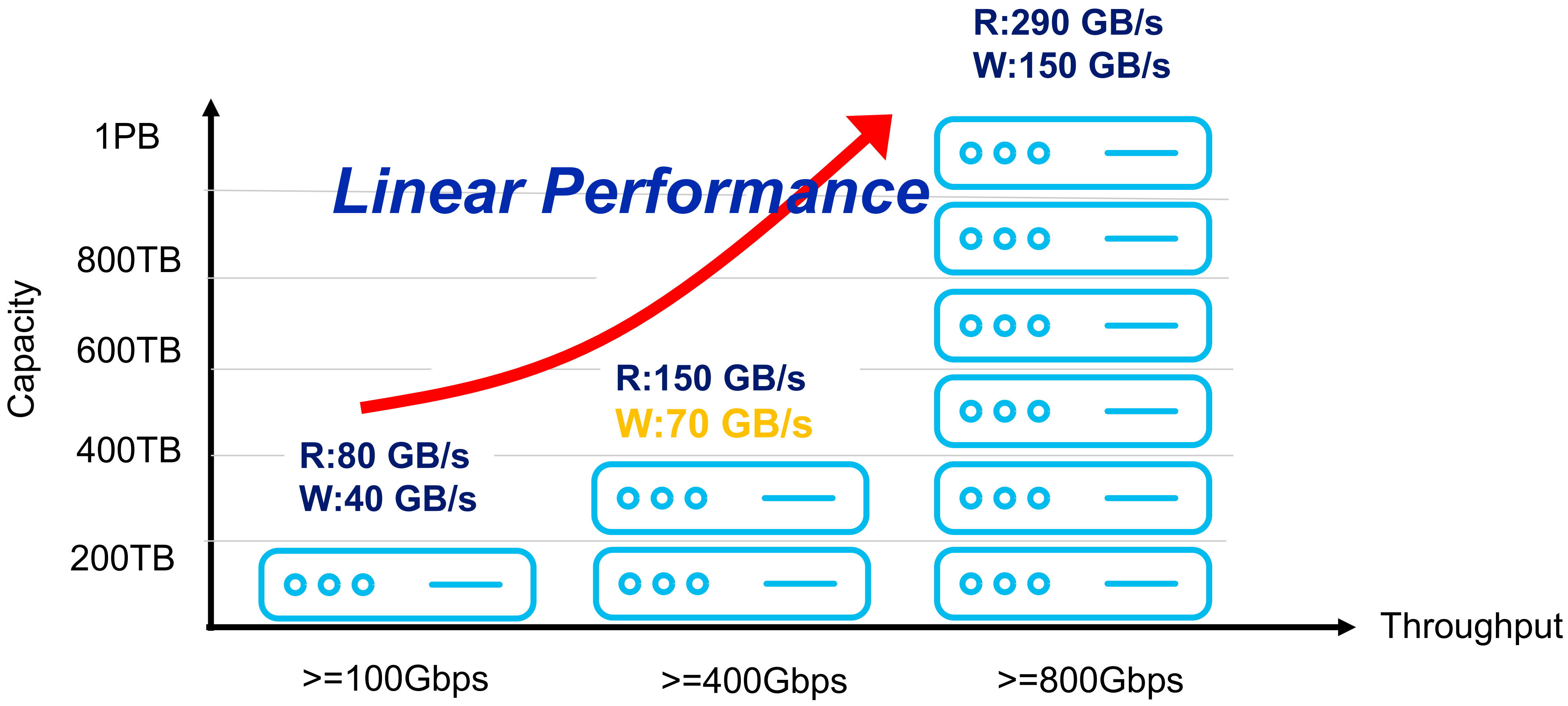
Unlimited Scaling up

By utilizing SupremeRAID™, single machine provides ultra-high throughput and easily scaling up with increasing data demands.

Native CSI Driver

Supports the latest CSI specifications for seamless integration with modern container infrastructure, ensuring flexible deployment and scalability.

From TB to PB: Flexible to Scaling out and up



Storage SuperServer SSG-221E-DN2R24R

2U all-flash SBB with 24 U.2 NVMe drives and PCIe 5.0

Key Applications

Artificial Intelligence (AI), High Availability Storage, NVMe Over Fabrics Solution

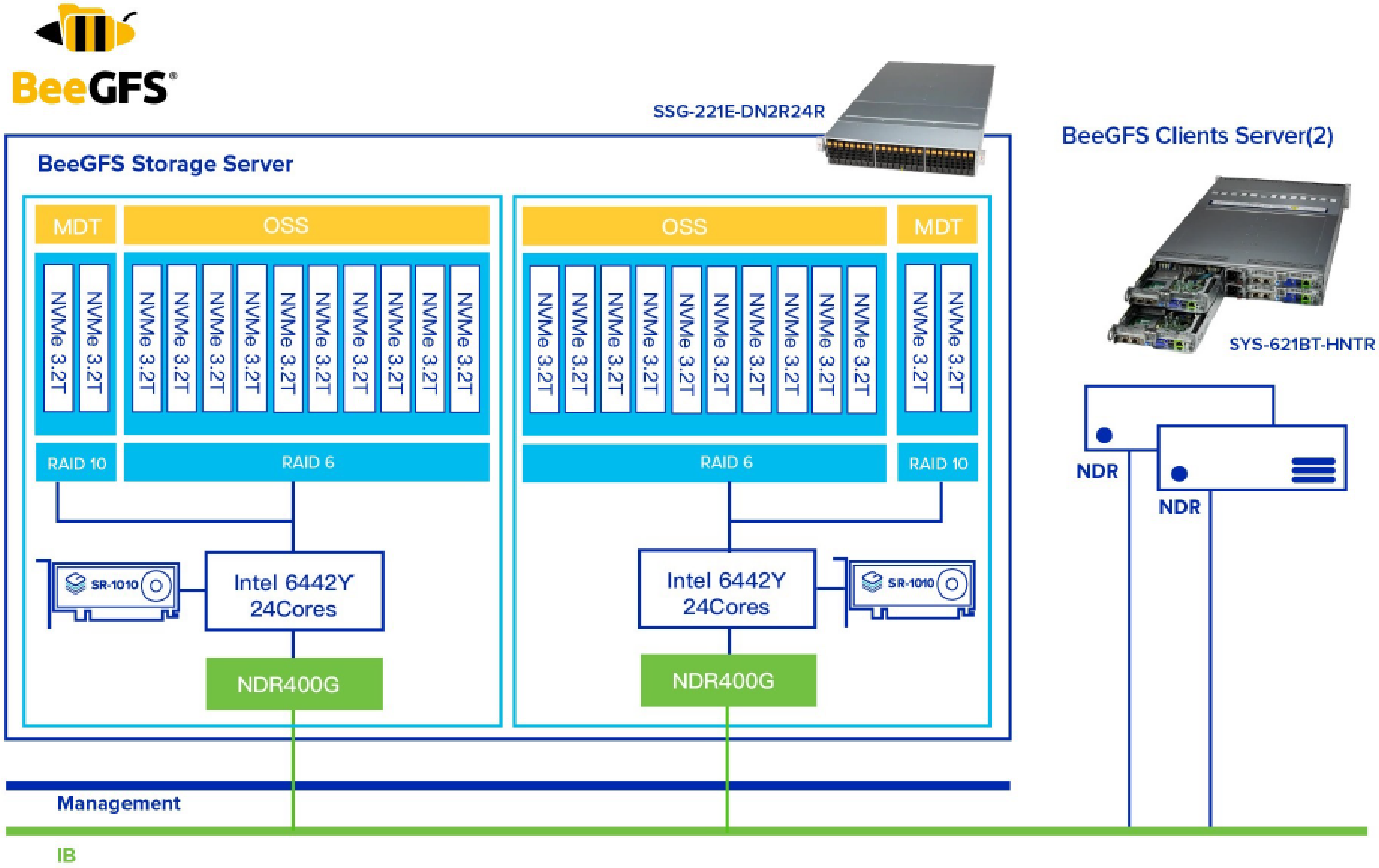
Key Features

Two hot-pluggable systems (nodes) in a 2U form factor. Each node supports the following:

- Single Socket E (LGA-4677) 5th/4th Generation Intel® Xeon® Scalable processor. Up to 350W TDP.
- 2 PCIe 5.0 x16 HHHL slots
  2 PCIe 5.0 x8 HHHL slots
- Supports 8 DIMMs per node with 1DPC, up to 2TB memory capacity with 8 DIMMs of 256GB 3DS RDIMM DDR5-5600 ECC memory per node.
- Dedicated 1GbE Private Ethernet connection for node-to-node communication (Heartbeat)
- 24 Hot-swap Dual Port Gen 5 U.2 NVMe SSD bays
- Redundant Titanium 2000W Power Supplies

SSG-221E-DN2R24R

Architecture in 2 Storage Nodes with 400G IB per each

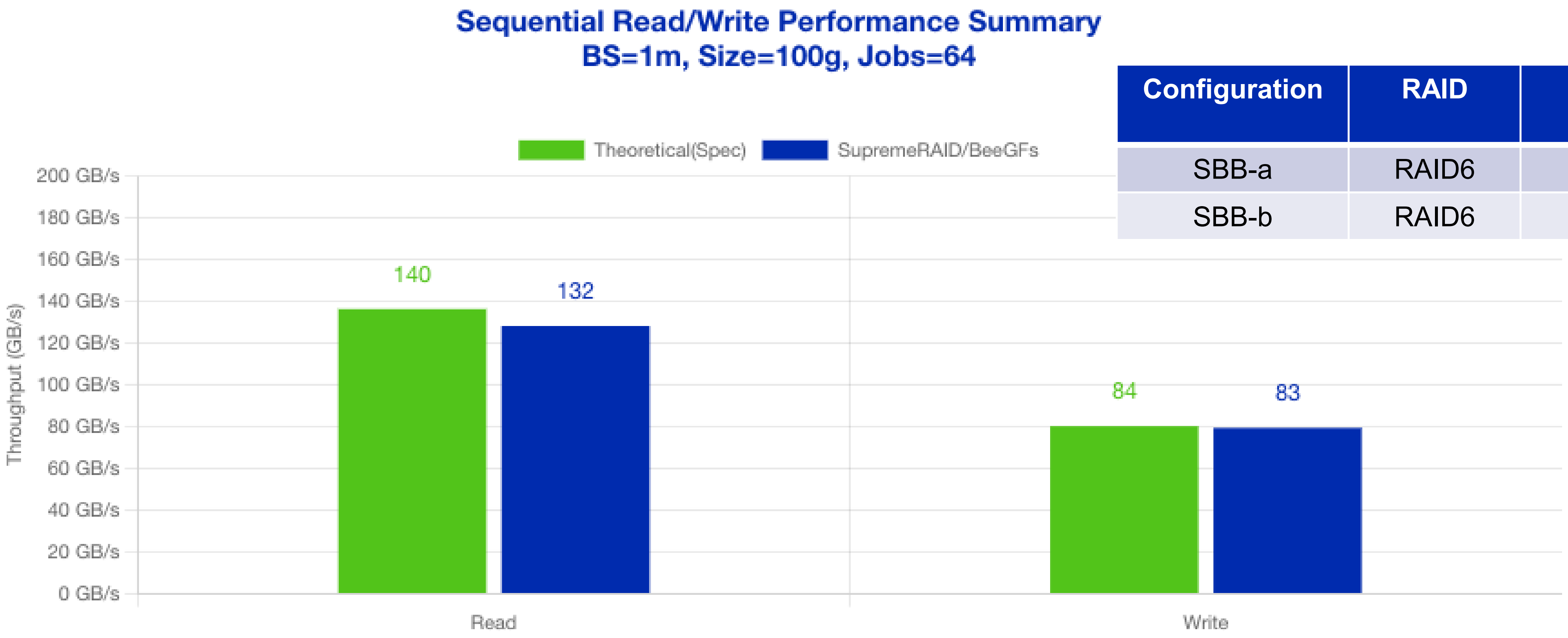


BeeGFS server configuration

Item	Storage server Specification	Per Node	Total Count	Client server Specification	Per Node	Total Count
Server	SuperMicro Storage SuperServer SSG-24E-NR24R (2U2NSP)	1	2	SuperMicro SYS-621BT-HNC8R (2U4NDP)	1	1
CPU	Intel(R) Xeon(R) Gold 6442Y	1	4	Intel(R) Xeon(R) Gold 6442Y	2	8
Memory	SK Hynix HMCG88AGBRA190N DDR5 32GB 5600MHz	8	32	Micron MTC10F1084S1RC48BA DDR5 16GB 4800MHz	16	64
SSD	SAMSUNG MZTL23T8HCLS-00A07 3.84TB Kioxia CM7-V 3.84TB	21 24+3	24 27	N/A	-	-
RAID Card	SR-1010	1	4	N/A	-	-
OFED Network	Mellanox MT2910 [ConnectX-7] NDR	1	4	Mellanox MT2910 [ConnectX-7] NDR	1	4
OFED version	5.8.5.1.1.2			5.8.5.1.1.2		
Operating System	RHEL9.4			RHEL9.4		
BeeGFS version	7.4.5			7.4.5		
InfiniBand Switch	Nvidia NVIDIA Quantum-2 InfiniBand QM9700					

One SBB/ 2 Nodes

BeeGFS storage server performance

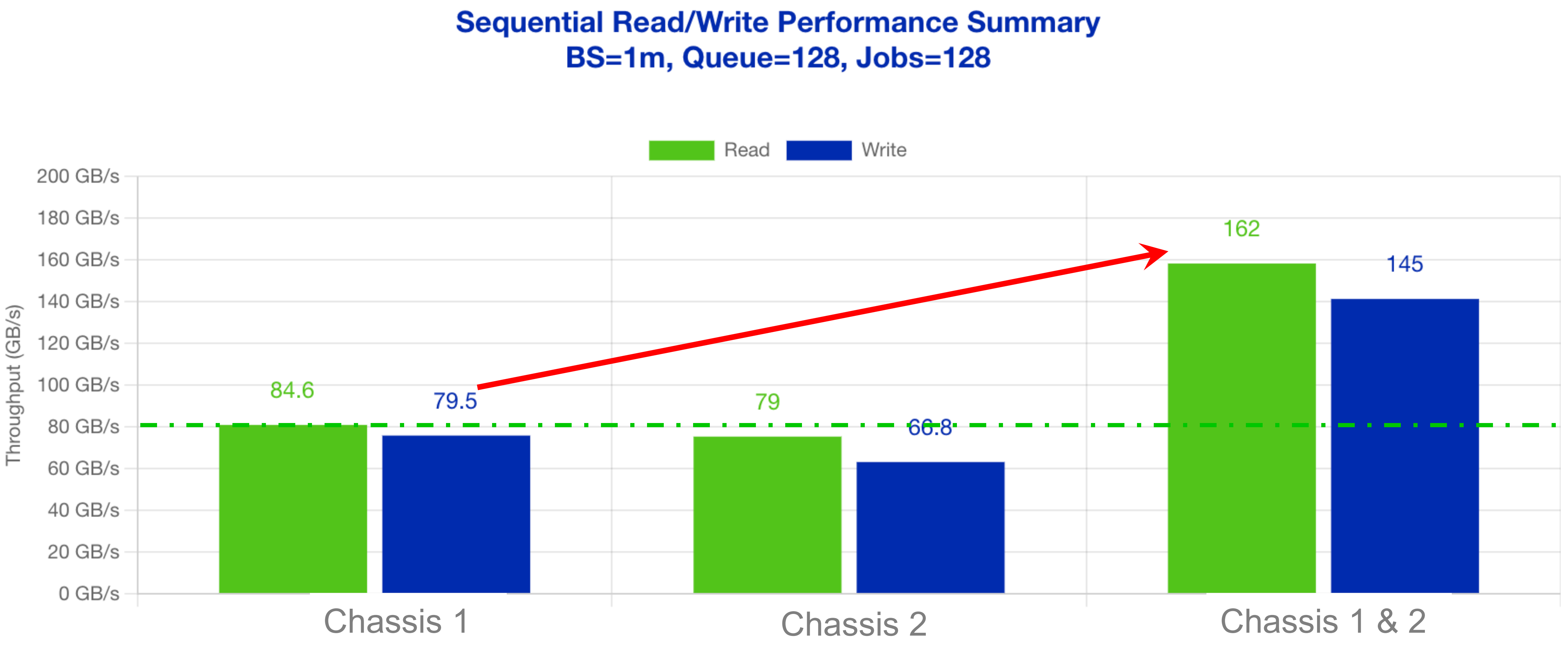


Configuration	RAID	NVMe Counts
SBB-a	RAID6	10
SBB-b	RAID6	10

	Spec	Result
Read(GB/s)	140	132
Write(GB/s)	84	83

2 SBB/ 4 Nodes

SBB Scale-out Performance



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The Best Cluster Solution for AI / HPC

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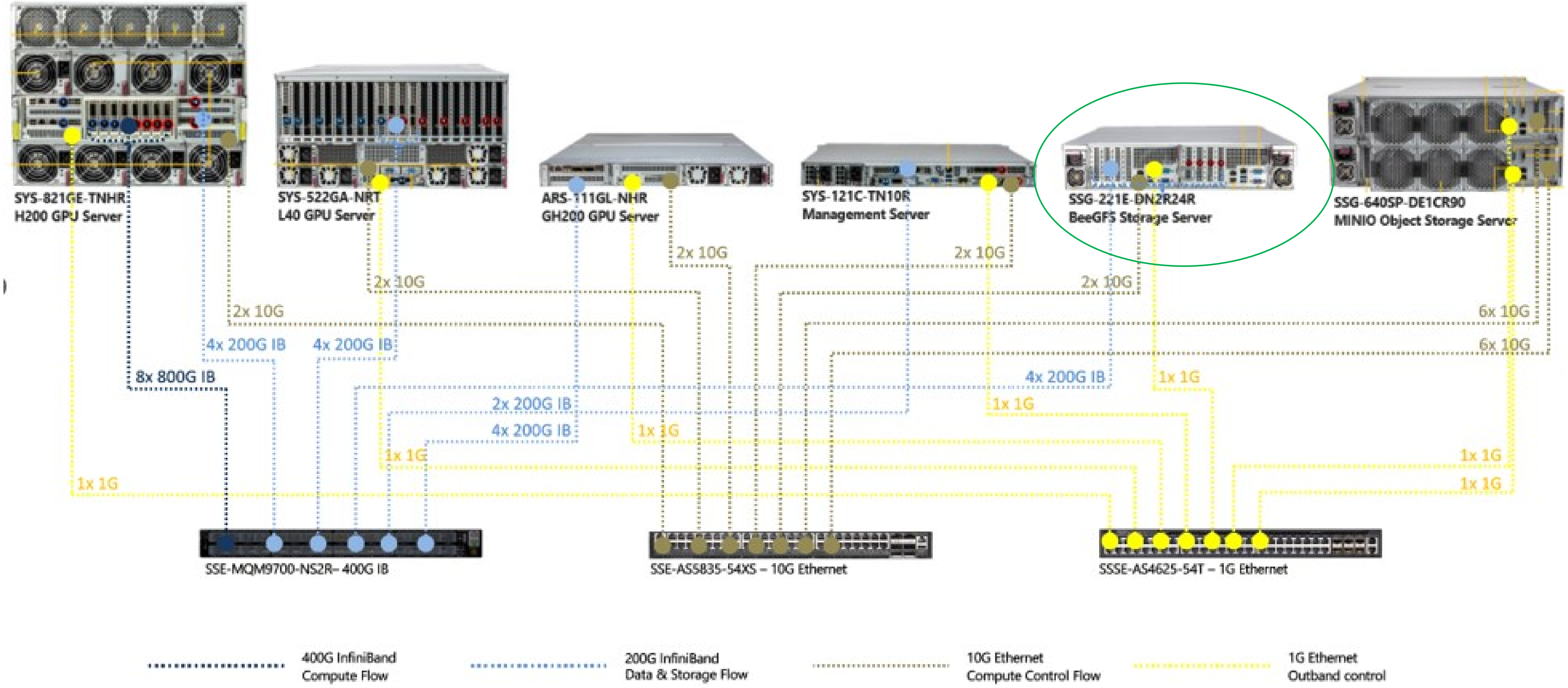
SMCI AI Topology



1 or 2 SBB to offer best performance for 2~32 HGX

Performance Characteristic	Good (GBps)	Better (GBps)	Best (GBps)
Single-node read	4	8	40
Single-node write	2	4	20
Single SU aggregate system read	15	40	125
Single SU aggregate system write	7	20	62
4 SU aggregate system read	60	160	500
4 SU aggregate system write	30	80	250

AI SuperPOD: The best Turnkey solution team

Data Scientist/
Developer/
AI Researcher

Generative AI

INFINITIX

AI-Stack

Self- Service, Scalable Infrastructure	Data Science& AI Workload Orchestration	Reproducibility & Collaboration	Governed Utilization of GPU Resource	Infrastructure Optimization	Cloud-Native Deployment	AI and Data Science Tolls and Frameworks

Container Orchestration and Management Integration

kubernetes

IT Administrator

Mainstream GPU Accelerated Servers

NVIDIA HGX

SUPERMICRO

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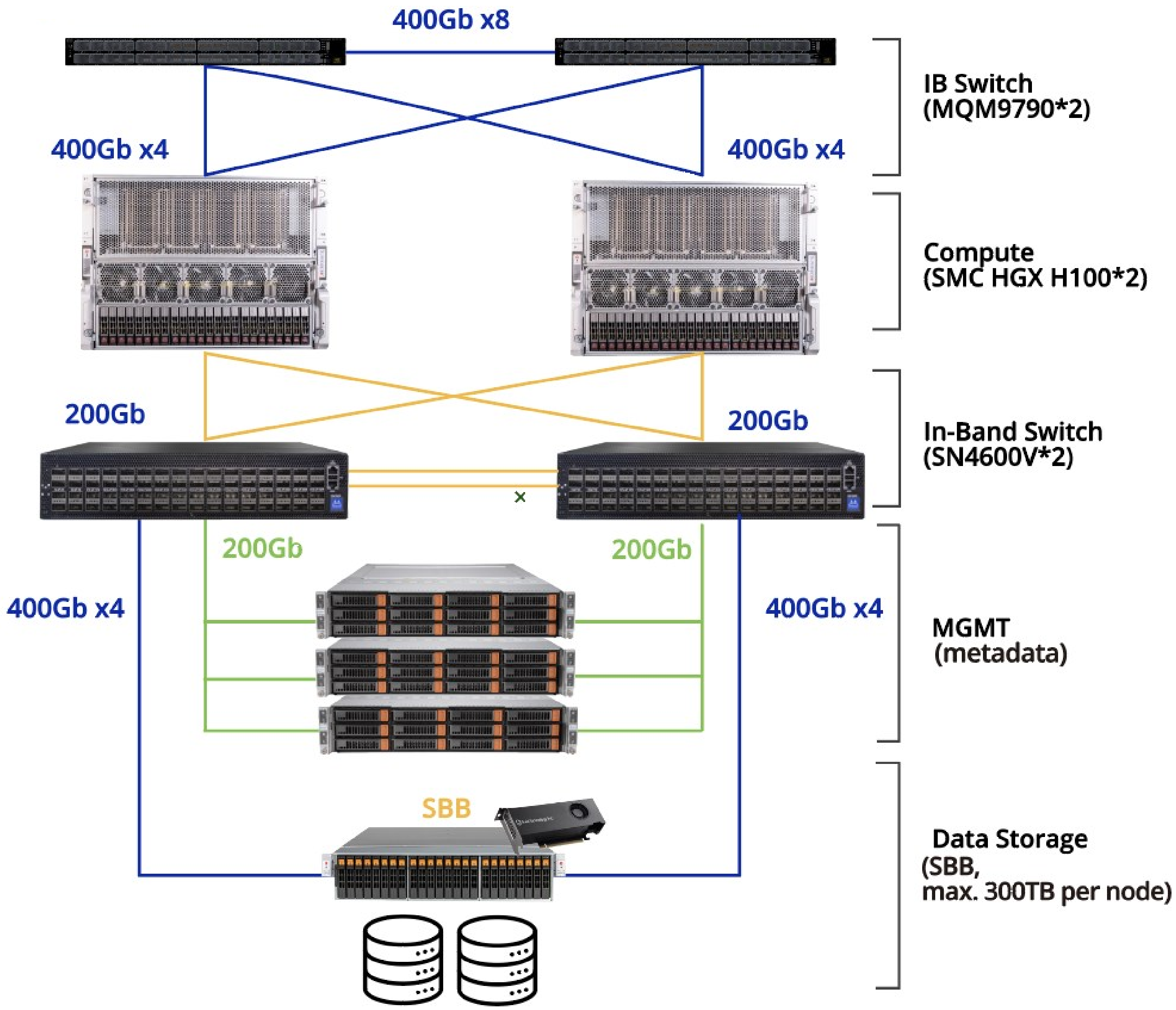
Accelerated Storages

BeeGFS

MINIO

Turning your SuperPOD into a real opportunity



WEKA / DDN/ SupremeStore Comparision

Feature	Weka.IO	DDN AI400X2 Turbo	SBB+SupremeRAID 1010+BeeGFS
Primary Function	High-performance parallel file system	High-performance AI storage	High-performance parallel file system (POSIX)
Primary Benefit	Unified namespace, high performance, and scalability	High-speed data access and processing for AI applications	Hihgest data performance and processing for AI/HPC with best price
Use Case	AI/ML, HPC, Financial Analytics, Life Sciences	AI/ML workloads, Data analytics, High-performance computing	HPC, AI & ML, Media & Entertainment, Life Sciences, and Oil & Gas…
Storage Type	NVMe (with integrated tiering to object storage)	SSD/NVMe	SSD/NVMe
Scalability out	Yes, highly scalable	Yes, highly scalable	Yes, highly scalable
Scalability up	Limited, releated on cpu performance	Limited, releated on cpu performance	Linear increase in performance with added SSDs
Performance	Good performance with low latency	Higher performance, lower latency	Extremely High performance, lowest latency
Performance	123 GiB/s (r), 37.6 GiB/s (w), 4M IOPs 1	115 GiB/s (r), 75 GiB/s (w), 3M IOPs 2	132 GiB/s (r), 83 GiB/s (w), 3M IOPs
Cost	Highest	Higher	Aggressive
Minimum Node Configuration	Typically 8 nodes (optimal performance)	Typically 2-4 nodes (exact number depends on deployment scale)	Flexible in 1 SBB nodes with Raid protection and high availabilty
Cloud Integration	Yes, with cloud-bursting and remote snapshots	X	Yes, BeeOND
Updates	Mannual and Complicted	Manual	Manual and Easy
Automatic Updates	X	X	X
Management Interface	Weka Home (Web-based management)	Web-based management	BeeGFS(CLI), SupremeRAID(CLI and Web-based)
Monitoring and Reporting	Weka Home (Advanced monitoring and reporting)	Advanced monitoring and reporting capabilities	Advanced monitoring and reporting capabilities
Configuration	Advanced configuration with dynamic rebalancing	Advanced configuration options	Advanced configuration options
Data Protection Method	Erasure Coding (EC)	CPU SW RAID	GPU RAID
High Availability	Erasure Coding (EC)	Replication	SBB Dual Controller + SupremeRAID cross node Fail-over
Support and Maintenance	Vendor-specific support	Vendor-specific support	Vendor-specific support
Snapshots	Local and remote snapshots	X	X
Volume Copy	X	X	X
Object Storage (S3)	Yes, with integrated tiering	X	X
Thin Provisioning	X	X	X
GPU Direct Storage (GDS)	Yes	Yes	Yes
Failover Time	X	<5 mins	20-60 Seconds